

## MAGONG, Taiwan

WMO: 467340

Lat: 23.569N

Lon: 119.628E

Elev: 103

StdP: 14.64

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 53.2       | 55.0 | 46.2                          | 46.5 | 55.2 | 48.3 | 50.2 | 56.3 | 37.8                  | 56.3 | 34.0 | 58.5 | 25.2                  | 20   | 0.575 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.5                    | 89.9            | 81.8 | 89.3 | 81.7 | 88.0 | 81.4 | 83.6                | 87.9 | 82.8 | 87.1 | 82.2 | 86.5 | 11.3                 | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 82.5                            | 170.0 | 87.1 | 81.5 | 164.4 | 86.2 | 80.8 | 160.7 | 85.5 | 47.9          | 88.1 | 46.9 | 87.9 | 46.3 | 87.3 | 87.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |  |
| (4) 36.2          | 31.9 | 29.6 | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |      |  |
| (5)               |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |      |  |
|                   |      |      | WB                         | 44.8 | 85.5               | 4.6 | 0.9                                                | 41.5 | 86.1       | 38.8 | 86.6       | 36.2 | 87.1       | 32.9 | 87.7 |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 74.0   | 63.1  | 62.9  | 66.7  | 73.0  | 77.9  | 81.5  | 83.5  | 83.1  | 81.5  | 76.9  | 71.5  | 66.1 |
|                                                                                  | DBStd       | 8.27   | 4.02  | 4.72  | 5.11  | 4.87  | 3.29  | 2.81  | 1.72  | 1.71  | 2.31  | 2.74  | 3.66  | 3.65 |
|                                                                                  | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | HDD65       | 252    | 84    | 90    | 40    | 5     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 30   |
|                                                                                  | CDD50       | 8773   | 407   | 361   | 517   | 690   | 865   | 945   | 1038  | 1025  | 945   | 835   | 645   | 500  |
|                                                                                  | CDD65       | 3550   | 26    | 31    | 92    | 245   | 400   | 495   | 573   | 560   | 495   | 370   | 198   | 65   |
|                                                                                  | CDH74       | 30480  | 33    | 64    | 246   | 1202  | 2985  | 5138  | 6757  | 6402  | 5009  | 2153  | 440   | 51   |
|                                                                                  | CDH80       | 8682   | 0     | 1     | 20    | 175   | 631   | 1567  | 2501  | 2177  | 1346  | 240   | 22    | 2    |
| Wind                                                                             | WSAvg       | 15.1   | 19.5  | 18.2  | 15.7  | 12.7  | 10.6  | 10.8  | 9.6   | 9.9   | 12.7  | 20.2  | 21.2  | 20.6 |
| Precipitation                                                                    | PrecAvg     | 37.4   | 0.8   | 1.9   | 2.2   | 3.3   | 4.6   | 6.6   | 5.2   | 6.6   | 3.4   | 1.1   | 0.9   | 0.8  |
|                                                                                  | PrecMax     | 57.4   | 3.3   | 12.4  | 10.6  | 17.2  | 10.7  | 13.7  | 16.7  | 25.2  | 15.7  | 8.1   | 5.4   | 3.3  |
|                                                                                  | PrecMin     | 20.9   | 0.1   | 0.0   | 0.1   | 0.2   | 0.1   | 0.4   | 0.2   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 11.5   | 0.7   | 2.4   | 2.2   | 3.2   | 3.0   | 4.3   | 4.4   | 5.6   | 3.7   | 2.0   | 1.1   | 0.9  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 78.3  | 79.1  | 82.4  | 85.7  | 89.4  | 90.6  | 91.6  | 91.2  | 89.9  | 86.2  | 82.5  | 78.9 |
|                                                                                  |             | MCWB   | 71.3  | 72.9  | 75.1  | 78.4  | 80.7  | 82.1  | 82.8  | 81.0  | 81.9  | 78.7  | 75.8  | 71.5 |
|                                                                                  | 2%          | DB     | 73.5  | 75.6  | 79.7  | 84.0  | 86.8  | 89.3  | 89.9  | 89.6  | 88.2  | 84.1  | 80.0  | 75.0 |
|                                                                                  |             | MCWB   | 68.1  | 70.7  | 74.3  | 77.3  | 80.1  | 81.9  | 82.2  | 81.4  | 81.2  | 77.8  | 74.4  | 70.0 |
|                                                                                  | 5%          | DB     | 70.2  | 72.8  | 77.0  | 82.3  | 85.2  | 87.9  | 89.3  | 88.2  | 87.4  | 82.5  | 78.2  | 73.0 |
|                                                                                  |             | MCWB   | 65.6  | 68.6  | 72.7  | 76.6  | 79.3  | 81.8  | 82.0  | 81.2  | 80.8  | 76.7  | 73.2  | 68.3 |
|                                                                                  | 10%         | DB     | 68.3  | 69.7  | 74.0  | 80.3  | 83.8  | 86.6  | 88.0  | 87.7  | 85.9  | 80.9  | 76.6  | 71.3 |
|                                                                                  |             | MCWB   | 64.2  | 66.2  | 70.7  | 75.4  | 78.5  | 81.2  | 81.7  | 81.1  | 80.0  | 75.4  | 72.3  | 66.7 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 71.8  | 75.0  | 76.9  | 79.7  | 82.3  | 84.2  | 84.6  | 84.6  | 83.7  | 80.1  | 77.1  | 73.4 |
|                                                                                  |             | MCDB   | 76.6  | 78.5  | 80.1  | 83.5  | 85.8  | 88.4  | 89.1  | 88.3  | 88.1  | 83.2  | 81.3  | 76.9 |
|                                                                                  | 2%          | WB     | 68.7  | 71.6  | 74.6  | 78.4  | 80.8  | 83.1  | 83.6  | 83.0  | 82.5  | 79.0  | 75.3  | 70.3 |
|                                                                                  |             | MCDB   | 72.5  | 75.1  | 78.8  | 82.2  | 85.2  | 87.3  | 88.2  | 86.9  | 86.6  | 82.3  | 78.8  | 73.8 |
|                                                                                  | 5%          | WB     | 66.4  | 69.0  | 72.8  | 77.2  | 79.8  | 82.3  | 82.8  | 82.3  | 81.5  | 78.0  | 74.2  | 68.8 |
|                                                                                  |             | MCDB   | 69.9  | 72.2  | 76.5  | 81.2  | 83.8  | 86.5  | 87.4  | 86.3  | 85.4  | 81.3  | 77.3  | 72.3 |
|                                                                                  | 10%         | WB     | 64.9  | 66.3  | 70.9  | 75.9  | 78.9  | 81.4  | 82.1  | 81.5  | 80.6  | 76.9  | 72.9  | 67.2 |
|                                                                                  |             | MCDB   | 68.2  | 69.6  | 73.9  | 79.6  | 82.4  | 85.4  | 86.7  | 85.6  | 84.3  | 80.0  | 75.9  | 70.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.3   | 8.7   | 9.3   | 9.5   | 9.4   | 9.2   | 9.5   | 9.0   | 8.7   | 7.9   | 7.8   | 7.8  |
|                                                                                  |             | MCDBR  | 11.1  | 13.5  | 13.7  | 11.7  | 10.9  | 10.3  | 10.7  | 10.1  | 10.4  | 9.1   | 9.6   | 10.0 |
|                                                                                  |             | MCWBR  | 7.9   | 9.8   | 10.0  | 7.4   | 6.7   | 6.1   | 5.9   | 5.6   | 6.1   | 5.9   | 6.6   | 6.6  |
|                                                                                  | 5% WB       | MCDBR  | 11.1  | 12.8  | 13.1  | 11.0  | 10.2  | 10.0  | 9.9   | 9.5   | 9.6   | 8.6   | 9.0   | 9.5  |
|                                                                                  |             | MCWBR  | 8.3   | 9.9   | 10.2  | 7.6   | 6.9   | 6.7   | 6.1   | 5.9   | 6.3   | 6.1   | 6.6   | 6.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.545  | 0.600 | 0.708 | 0.698 | 0.569 | 0.513 | 0.516 | 0.560 | 0.591 | 0.588 | 0.553 | 0.553 |      |
|                                                                                  | taud        | 1.894  | 1.791 | 1.608 | 1.672 | 2.008 | 2.200 | 2.183 | 2.037 | 1.921 | 1.873 | 1.942 | 1.889 |      |
|                                                                                  | Ebn at Noon | 224    | 221   | 205   | 211   | 239   | 251   | 251   | 240   | 229   | 221   | 220   | 215   |      |
|                                                                                  | Edh at Noon | 56     | 67    | 84    | 80    | 57    | 47    | 47    | 55    | 60    | 61    | 54    | 55    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 874    | 1001  | 1197  | 1397  | 1616  | 1750  | 1960  | 1736  | 1639  | 1397  | 1018  | 879   |      |
|                                                                                  | RadStd      | 138    | 166   | 140   | 179   | 208   | 177   | 139   | 204   | 152   | 130   | 130   | 106   |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (20 neighbors) | +0.50 | N/A     | N/A    | +0.36   | +0.45 | +0.45 | N/A         | N/A   | +229  | +214  |

Nomenclature: See separate page